

Mechman 48V Alternator + WS500 + APM-48 Install Guide

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Purpose: one shop-reference document for installing and commissioning the Hiatus dedicated 48V secondary alternator path: Mechman 48V alternator/bracket, Wakespeed WS500 , Balmar APM-48 , Ford Upfitter #3 enable, and the existing 48V house bank/Lynx architecture.

This is an implementation guide, not a substitute for the official manuals. Use the official PDFs for diagrams, connector pinouts, belt routing images, torque values, and firmware/app screens. This guide owns the build-specific sequence, gates, and no-go rules.

Current build baseline

- Truck: 2021 Ford F-350 7.3L gas / Godzilla platform.
- Alternator path: dedicated 48V secondary alternator, regulated by Wakespeed WS500 .
- Protection layer: Balmar APM-48 at the alternator.
- House bank: 3x Dumfume 51.2V 100Ah LiFePO4 in parallel (1S3P), internal BMS, no confirmed CAN/closed-loop Wakespeed integration.
- House distribution: Victron Lynx Distributor, SmartShunt, MultiPlus-II 48/3000/35-50 , Orion-Tr Smart 48/12-30 .
- Manual charge enable: Ford Upfitter Switch #3 -> F-15 3A inline fuse -> WS500 brown ignition/enable wire .
- Main alternator branch: 2/0 AWG positive through F-04 200A/80V MEGA at Lynx Slot 3 / house-bank end; dedicated 2/0 AWG negative direct to Lynx/system negative. The Wakespeed 500A/50mV shunt sits on the battery side of the SmartShunt in the common battery-negative path; the SmartShunt remains hard-attached to the Lynx.
- Current closeout report (2026-09-02): alternator charging remains disabled/not yet connected as an active charge source. The owner reports only two remaining physical/configuration items: wire the protected PH-VAN red lead and read/save/program the WS500 through the purchased USB A-to-B/printer-style cable. Treat the full preflight below as a verification pass over the as-built hardware—not a claim that every older unchecked inventory line is still undone. Do not energize field or perform first charge until the PH-VAN feed, saved/programmed profile, sensor/APM/shunt/enable/fuse topology, and shutdown-response gates below pass.

Current physical/staged-driving status

Owner update, 2026-06-11 : the alternator noise was traced to the Mechman-supplied idler pulley not being seated properly. After tightening/seating the idler, the noise went away. Mechman also confirmed that the truck can be driven with the 48V alternator mechanically installed and unwired/electrically disabled.

Practical staged-install status:

1. **Mechanical-only staged driving is acceptable for this build after the belt/idler/noise check passes, but it is still a disabled alternator, not a commissioned charging system.**
2. The alternator must remain electrically inert while unwired:
 - WS500 not enabled;
 - field/regulator connector disconnected or capped/isolated;
 - alternator B+ booted/capped and not connected to an unterminated live cable;
 - all studs/cable ends insulated and strain-relieved;
 - no partial wiring can accidentally energize the field.
3. Do **not** drive with the alternator capable of producing power into an unfused, unterminated, partially wired, or disconnected house-bank path.
4. Do **not** energize the WS500 / field circuit until main charge cable, dedicated negative return, APM-48, fusing, sense wiring, shunt/current sense, temperature sensing, and WS500 profile are installed and checked.
5. If belt/idler noise returns, treat it as a mechanical belt/pulley seating issue first: stop, inspect alignment/tension/hardware, then re-test at idle before road use.

Staged approach:

- Stage 1: bracket/alternator/belt installed; alternator electrically inert; idler seated; belt/noise check passed; road driving OK in this disabled state.
- Stage 2: install/route large cables, APM, and rough-in WS500 wiring, but keep WS500 disabled until termination/protection is complete.
- Stage 3: wire/configure WS500 with Upfitter #3 OFF; verify app/profile/sensors before first enable.
- Stage 4: first charging run with meters and a planned shutdown path.

Hard no-go rules

- Do not allow the 48V alternator to charge without a properly configured WS500.

- Do not open the 48V battery/main disconnect while the engine is running and the alternator is charging.
- Do not use the main 48V disconnect as the normal alternator shutdown method.
- Do not rely on the APM-48 as the primary shutdown method. It is a surge clamp / backup protection layer.
- Do not connect the alternator positive cable without source-side protection at the house-bank/Lynx end.
- Do not rely on chassis/sheet-metal/factory grounds as the normal 48V alternator return path.
- Do not bury the WS500, APM, fuses, or sense/shunt wiring where inspection is impossible.
- Do not intentionally provoke a battery BMS disconnect while the alternator is producing current.
- Do not assume Victron SmartShunt/Cerbo DVCC is a cell-level BMS or a replacement for Wakespeed/BMS communication.

Source documents reviewed

Local repo references

- references/Mechman General 48v install INSTR-48V-GEN_3-28-2025.pdf
- references/Mechman 48v Alt Install Guide INSTR-48V-GEN_3-28-2025.pdf — duplicate of the general 48V install guide.
- references/Mechman dual alternator bracket INSTR-GZDB-20.pdf
- media/references/mechman-ws500-apm48-install-guide/7.3l-single-generator-belt-routing-owner-confirmed-2026-06-08.jpg — owner-confirmed 7.3L single-generator belt-routing orientation reference.
- references/WS500-Product-Manual-09-30-2022-V2.pdf
- references/WS500-Quick-Start-Guide-09-30-2022-V3.pdf
- references/Alternator Protection Module PDS-APM-24.pdf — official Balmar APM quick-start PDF, despite filename; applies by module voltage class.
- references/Dunfume_36V_48V_100Ah_Battery_-_User_Manual.pdf

Online source URLs checked in this pass

- Mechman 48V Alternator Installation Instructions: <https://www.mechman.com/instructions/48v-alternator-installation-instructions/>
- Mechman INSTR-48V GEN_3-28-2025 PDF: https://mechman.com/content/instructional-pdfs/INSTR-48V%20GEN_3-28-2025.pdf
- Mechman 48-Volt Warning : <https://www.mechman.com/instructions/48volt-warning/>
- Mechman INSTR-48V-WARN_3-28-2025 PDF: https://mechman.com/content/instructional-pdfs/INSTR-48V-WARN_3-28-2025.pdf
- Mechman 7.3L Godzilla Dual Alternator Bracket instructions: <https://www.mechman.com/instructions/73l-godzilla-dual-alternator-bracket-assembly-instruction-manual/>
- Mechman INSTR-GZDB-20 PDF: <https://mechman.com/content/instructional-pdfs/INSTR-GZDB-20.pdf>
- Wakespeed WS500 product page: <https://www.wakespeed.com/product/ws500-advanced-alternator-regulator/>
- Wakespeed WS500 Product Manual 10.21.24 : <https://www.wakespeed.com/wp-content/uploads/WS500-Product-Manual-10.21.24-compressed.pdf>
- Wakespeed Communications and Configuration Guide v2.6.1 : <https://www.wakespeed.com/wp-content/uploads/Wakespeed-Communications-and-Configuration-Guide-v2.6.1-1.pdf>
- Wakespeed WS500 Data Sheet 9-26-22 : <https://www.wakespeed.com/wp-content/uploads/Wakespeed-WS500-Data-Sheet-9-26-22.pdf>
- Wakespeed Victron Cerbo GX Guide 4.29.24 : https://www.wakespeed.com/wp-content/uploads/Wakespeed-Victron-Cerbo-GX-Guide_4.29.24-1.pdf
- Balmar APM-48 product page: <https://balmar.net/product/apm-48/>
- Balmar Alternator Protection Modules page: <https://balmar.net/alternator-protection-modules/>
- Balmar APM quick-start / product sheet: <https://balmar.net/wp-content/uploads/2022/09/PDS-APM-24.pdf>

What the official docs say that matters

Mechman 48V general install guide

- Use pure copper cable.
- Aim to limit voltage drop to 2% between alternator and battery bank.
- General cable chart:
 - up to 100A , 5 ft one-way: 6 AWG ;
 - up to 150A , 10 ft one-way: 4 AWG ;
 - up to 200A , 15 ft one-way: 2 AWG ;
 - up to 250A , 20 ft one-way: 1/0 AWG .
- Verify sizing separately for runs longer than 20 ft or systems over 250A .
- Fuse the positive charge cable within 12 in of the battery-bank connection.
- Ground cable must be equal to or larger than the positive cable and must connect directly from alternator to battery bank.
- Avoid sheet metal/factory grounds for the 48V alternator return path.

- Do not operate the system without a properly configured regulator.
- Fully charge the 48V batteries with a compatible charger before first startup.
- First-start check: start vehicle, apply modest load, raise RPM to 2500 , verify bank voltage rises at least 1V , and verify charge and ground path voltage drop each show <0.1V under load.

Mechman 7.3L Godzilla bracket guide

- Kit contents include:
 - dual alternator bracket for Ford 7.3L ;
 - 7.3L step washer;
 - smooth idler pulley;
 - 88.06 in serpentine belt;
 - M10/M8 flanged hardware.
- Mechanical sequence:
 1. Attach bracket to engine with 4x M10-1.5 x 90mm flanged bolts.
 2. Install idler pulley with step washer and M8-1.25 x 35mm flanged bolt.
 3. Install OEM-compatible saddle-mount alternator with M10-1.5 x 90mm and M10-1.5 x 40mm bolts.
 4. Install second OEM-compatible T-mount alternator with 3x M10-1.5 x 80mm bolts.
 5. Install the supplied 88.06 in serpentine belt per the Mechman diagram.
- The guide text extraction does not include torque values. Use Ford/Mechman service data for torque, not guesses.

Owner-confirmed single-generator routing orientation, 2026-06-08 :

- On the truck's 7.3L single-generator belt layout, label A in the saved reference image is the belt span **closest to the engine**.
- Label B is the belt span **furthest from the engine**.
- Use this as an orientation aid while still following the official Mechman/Ford routing and torque instructions for the actual install state.

7.3L single-generator belt-routing orientation: A closest to engine, B furthest from engine

Wakespeed WS500 wiring/control facts

- Brown ignition/enable wire must see at least 8.5VDC to turn the WS500 on.
- Brown can be fed from ignition, oil-pressure switch, or another circuit active only when engine is running.
- In this build, brown is fed by Ford Upfitter #3 through F-15 3A .
- White Feature-In is configurable; in LiFePO4/CPE use it can force float or be assigned to other behavior. Do not assign it casually.
- Red alternator positive / regulator power wire: fuse at 10A , or 15A for extra-large-case alternator.
- Red/yellow positive battery voltage-sense wire: fuse at 3A .
- Black/yellow negative sense wire: connect to negative of the bank being charged.
- Purple/grey shunt-current-sense pair: default 500A/50mV . For the selected common battery-negative layout, purple/current-sense-high lands on the battery-negative side of the Wakespeed shunt and grey/current-sense-low lands on its SmartShunt/Lynx system side; no sense-wire fuses are used. Configure Shunt at Battery .
- Alternator temp sensor mounts to rear case bolt or alternator ground-terminal bolt.
- Battery temp sensor is important on LiFePO4 because charging must stop outside battery temperature limits.
- Power and voltage-sense wires must be fused and must be placed so the regulator does not lose the battery reference while the alternator is still operating.
- Wakespeed notes that the positive battery switch shown in its wiring diagrams must be ON whenever engine is running.

Wakespeed 48V / high-energy cautions

- Wakespeed treats 48V lithium alternator systems as high-energy systems where uncontrolled disconnect/load-dump can create destructive voltage spikes.
- Wakespeed documentation describes 48V load-dump spikes upward of 400V if the battery/contactors/FETs disconnect before the alternator can be shut down.
- Alternators may need hundreds of milliseconds to shut down; Wakespeed guidance discusses needing advance notice before disconnect, with 2 seconds as a practical warning target in their high-energy discussion.
- CAN/BMS integration is materially safer than a simple charge-enable wire because it can communicate limits and fault state before disconnect.
- DVCC/Cerbo monitoring is not a replacement for true WS500/BMS safety communication.
- The Dumfume bank has internal BMSs and no confirmed CAN integration, so treat BMS disconnect as an emergency backstop, not a normal charge-control event.

Balmar APM-48 facts

- APM-48 is a surge/load-dump protection module for 48V alternator systems.
- Install at the alternator, across alternator output/ground points:
 - red APM ring terminal to alternator B+ post;
 - black APM lead to alternator B- if isolated ground, or clean alternator case mounting bolt if case ground.
- Secure the APM case to a battery cable with zip ties.
- Do **not** place either APM connector under the battery cable lugs.
- Green LED means protecting.
- Red LED and/or beep/chirp means protection failure; replace APM.
- If alternator cable nuts loosen, the APM cannot protect the alternator.
- No surge protection device can protect against unlimited surge amplitude, power, or duration.

System map for this build

High-current path

1. Alternator B+ output stud.
2. 2/0 AWG positive charge cable routed to house electrical board.
3. F-04 200A/80V MEGA in Lynx Slot 3 near the house-bank/Lynx end.
4. Lynx positive bus.
5. Battery bank through main disconnect/battery-side protection.

The APM-48 is **not** a series device in that path. It is a parallel surge clamp installed at the alternator between B+ and B- /case ground.

Return path

1. Alternator B- / approved negative terminal / approved ground point.
2. Dedicated 2/0 AWG negative return cable directly to Lynx/system negative; do not use chassis-only return.
3. All battery current then passes through Lynx/system negative -> Victron SmartShunt -> Wakespeed 500A/50mV shunt -> battery negative combine .

Mount the Wakespeed shunt next to, but on the battery side of, the SmartShunt in the common battery-negative path: battery negative combine -> Wakespeed shunt -> SmartShunt -> Lynx . Leave the SmartShunt physically hard-attached to the Lynx. Purple/current-sense-high faces battery negative; grey/current-sense-low faces the SmartShunt/Lynx system side. No load or charge return may land on the battery side of either shunt.

Regulator/control path

1. Ford Upfitter #3 factory switched 12V output.
2. F-15 3A inline fuse close to the upfitter/control-wire source.
3. 16 AWG TXL/GXL control lead to WS500 brown ignition/enable wire.
4. Upfitter #3 ON allows WS500 to run; Upfitter #3 OFF disables WS500 field control.

WS500 PH-VAN sense/protection path

- The confirmed PH-VAN harness joins regulator power and positive voltage sense on one short red lead; do not extend the harness itself.
- Feed that red lead from the alternator/load side of F-04 through one Eaton/Bussmann HEB-AA in-line holder and Littelfuse KLKD015.T 15A/600VAC/DC fast-acting midjet fuse (F-12/F-13-PHVAN , active BOM row 171). Secure the single holder in the loom immediately beside F-04 ; use short 14 AWG pigtails and one sealed 14-to-16 AWG splice. Do not add a DC fuse panel, DIN rail, or enclosure, and do not use the incomplete Littelfuse housings or existing 20A fuses.
- Purple/grey current-sense pair runs unfused to the Wakespeed shunt in the common battery-negative path: purple/high on the battery-negative side and grey/low on the SmartShunt/Lynx system side.
- Alternator and battery temperature sensors are installed before first charging run.

Installation phases

Phase 0 — preflight and documentation at the bench

Before opening the truck:

- Print or save locally:
 - Mechman INSTR-GZDB-20 bracket PDF for belt routing image;
 - Mechman INSTR-48V GEN_3-28-2025 ;
 - Mechman 48V warning sheet;
 - Wakespeed WS500 current product manual;

- Wakespeed communications/configuration guide;
- Balmar APM quick-start sheet.
- Photograph the current belt path and engine-bay state before disassembly.
- Confirm Mechman alternator terminals/labels:
 - B+ output;
 - B- / negative / case-ground behavior;
 - field terminal/harness plug;
 - stator/tach terminal if used;
 - temp sensor mounting point.
- Confirm WS500 harness type matches alternator field polarity:
 - WS500-PH for positive/B-type field;
 - WS500-NH for negative/A-type field.
- Confirm whether Mechman's supplied alternator field is 12V or true 48V . Many 48V alternators use a 12V field; Wakespeed has derate guidance for this. Do not guess.
- Confirm all required fuses/holders are present and voltage-rated for the circuit they are on.
- Confirm the batteries are above low-temp charge cutoff and not in any BMS protection state.

No-go if these are unknown:

- field polarity / harness type;
- alternator ground style;
- whether a terminal is B+ , B- , field, or stator;
- WS500 profile/configuration basis;
- battery temperature state.

Phase 1 — mechanical bracket/alternator install

Official Mechman order:

1. Engine cool; ignition OFF.
2. Disconnect negative terminals of all batteries involved in the work area.
3. Install dual alternator bracket with 4x M10-1.5 x 90mm bolts.
4. Install smooth idler pulley with step washer and M8-1.25 x 35mm bolt.
5. Install saddle-mount alternator with M10-1.5 x 90mm and M10-1.5 x 40mm bolts.
6. Install T-mount alternator with 3x M10-1.5 x 80mm bolts.
7. Install 88.06 in belt using Mechman belt-routing diagram.
8. Check pulley alignment, belt tension, clearance, and no-contact zones.
9. Torque hardware to OEM/bracket specs.

Build-specific mechanical checks:

- Rotate/check belt path visually before starting.
- Confirm no harness, coolant line, AC line, or loom touches belt/pulley/fan path.
- Confirm alternator output stud cannot touch bracket/hood/pipework even under vibration.
- Confirm APM and wiring can be added without removing the alternator again, if possible.

If stopping here and driving before wiring:

- leave the alternator electrically inert;
- cap/cover output and field terminals;
- verify no harness can energize field;
- idle-check before road driving;
- recheck belt alignment/noise after the first short drive.

Phase 2 — APM-48 and high-current cable routing

Install the APM at the alternator before any live charging use.

APM steps:

1. Identify alternator B+ and B- / case-ground point.
2. Connect APM red ring to alternator B+ post.
3. Connect APM black:
 - to alternator B- post if isolated-ground;

- to clean alternator case mounting bolt if case-ground.
4. Do **not** place APM terminals under main battery cable lugs.
 5. Secure APM case to nearby battery cable with zip ties, away from belt/heat/abrasion.
 6. Verify APM green LED when system is live; red/chirp means replace.

High-current cable routing:

- Positive: alternator B+ to Lynx Slot 3 / F-04 200A/80V MEGA at house-bank end.
- Negative: alternator B- /approved ground directly to Lynx negative with dedicated 2/0 AWG . The Wakespeed shunt is not in this branch; it sits on the battery side of the SmartShunt in the common battery-negative path.
- Protect both cables from chafe, heat, steering/suspension movement, hood/engine movement, and sharp pass-throughs.
- Use grommets/bulkheads/loom/P-clamps/J-clamps where conductors pass through sheet metal or near edges.
- Keep unfused positive exposure as short as practical. Mechman says fuse within 12 in of battery-bank connection; in this build that protection is the house-end F-04 branch fuse.
- Do not leave a long alternator positive cable connected at one end and floating/unterminated at the other.

Field routing card, owner-confirmed route 2026-06-17 :

- Preferred physical route: alternator area -> fixed frame path under truck -> up at front of bed -> existing/front-bed-wall grommet -> electrical panel area.
- Route the two 2/0 AWG cables first because they dictate bend radius and support points; keep B+ and dedicated B- together on the same protected route.
- Protect the 2/0 with split loom, abrasion sleeve, rubber hose chafe guards, or equivalent wherever it contacts/approaches frame edges, brackets, pass-throughs, or zip-tie support points.
- Add strain relief before and after the bed-wall grommet; do not let cable movement saw against the pass-through. Use a gentle bend and a subtle drip loop outside the grommet if water can track along the cable.
- Use rubber-lined P-clamps/Adel clamps on existing bolts/holes where cleanly available. Heavy UV/heat-rated zip ties are acceptable as support on fixed frame structure if paired with chafe protection, but do not tie to brake lines, fuel lines, factory harnesses, steering, suspension, driveshaft-adjacent parts, or anything that moves.
- Zip-tie spacing target: about 12-18 in on straight runs, closer near bends, transitions, and bed entry. Do not cinch so tightly that the tie bites into insulation.
- Keep the already-loomed WS500 harness near the route but not tightly lashed to the 2/0 where avoidable. Focus extra protection on the large unfused/charge conductors and pass-throughs.

Voltage-drop target:

- Mechman target: 2% alternator-to-bank voltage drop.
- Existing build lock: use on-hand 2/0 AWG for the assumed ~20 ft one-way positive and dedicated negative return, giving strong margin for the published 145.7A output curve and F-04 200A cable-protection fuse.
- First-run measured target from Mechman: charge path and ground path each <0.1V drop under load.

Rough-in bundle to run while the alternator chase is open

Preferred WS500 placement for this build: mount the WS500 near the house electrical board / truck-bed battery and shunt area, not deep in the engine bay. This keeps the analog shunt-current and battery-sense wiring short and serviceable. Run the alternator-leg wiring forward to the Mechman unit.

Owner-confirmed harness finding, 2026-06-17 : the supplied Wakespeed harness is the PH-VAN style harness, not the older/basic harness assumed by some diagrams. Practical implications:

- The VAN harness has a single short red/black pair. Treat these as the combined WS500 regulator power and battery voltage-sense pair; land them at the house/main bus area near the WS500 with appropriate small fusing. Do **not** try to run those short red/black leads to the alternator.
- The VAN harness has no orange lamp/feature-out wire; absence of orange is normal.
- The short yellow/green connector is the CAN high/low connector. With the current Dumfume internal-BMS/no-confirmed-CAN setup, label/protect it as unused and do not connect it to alternator, battery, field, stator, or ground.
- The blue/yellow alternator connector uses the provided Mechman one-wire adapter: blue passes through as the field-control wire; yellow is the generic stator/AC/tach sense lead and is unused/dead-ended with this adapter unless Mechman/Wakespeed explicitly require a stator/tach signal.

Run these as separate, labeled, protected looms rather than one messy bundle:

1. High-current charge pair

- 2/0 AWG positive from alternator B+ to Lynx Slot 3 / F-04 200A/80V at the house end.

- 2/0 AWG dedicated negative return from alternator B- / approved case-ground point directly to Lynx negative.
- Keep these physically protected from heat, chafe, steering/suspension movement, and sharp pass-throughs.

2. WS500 alternator leg: engine bay ↔ WS500

- Blue field lead to the alternator field terminal through the provided Mechman one-wire adapter.
- Yellow stator/AC/tach lead is present in the blue/yellow connector but unused/dead-ended by the supplied adapter unless Mechman/Wakespeed explicitly require a stator/tach signal.
- The PH-VAN harness red/black pair does **not** run forward to the alternator in this build; keep it local at the WS500/house bus area as the combined regulator power and voltage-sense pair.

3. Alternator temperature sensor

- Mount at the alternator rear case bolt or ground-terminal bolt.
- If extended, keep at least 4 in from noise sources and use shielded instrument cable where practical, shield grounded at one end only.

4. WS500 battery/shunt/control leg: local near house bank

- PH-VAN red combined regulator power / positive voltage sense through the appropriate small fuse to the charged house/main positive bus.
- PH-VAN black combined regulator negative / negative voltage sense to the matching house/main negative bus.
- Purple/grey current-sense pair to the Wakespeed shunt on the battery side of the SmartShunt in the common battery-negative path; no inline fuses. Purple/current-sense-high lands on the battery-negative side and grey/current-sense-low lands on the SmartShunt/Lynx system side. Use twisted pair / instrument cable, keep routing quiet, configure Shunt at Battery , and verify positive charging-current sign before trusting readings.
- Battery temperature sensor at/near the house battery bank if used by the final profile.
- Yellow/green CAN connector labeled/protected as unused unless a compatible CAN/BMS integration is later added.

5. Cab/control leg

- Brown ignition/enable from Ford Upfitter #3 through F-15 3A to the WS500.
- White Feature-In is reserved for future charge-disable / force-float interlock work; rough in a spare control conductor if the chase is open, but do not depend on it for Phase 1.
- Orange lamp/alarm is optional; run only if a dash warning lamp/audible alarm is wanted.

Phase 3 — WS500 harness and control wiring

WS500 must be mounted, wired, and configured before charging is allowed.

Minimum required WS500 connections for this build with the confirmed PH-VAN harness:

- brown ignition/enable to Ford Upfitter #3 through F-15 3A ;
- PH-VAN red combined regulator power / positive voltage sense through the appropriate small fuse at the house/main positive bus;
- PH-VAN black combined regulator negative / negative voltage sense at the matching house/main negative bus;
- blue field lead through the provided Mechman one-wire adapter;
- yellow stator/tach lead remains unused/dead-ended unless Mechman/Wakespeed explicitly require it;
- purple/grey current-sense pair to the common battery-negative Wakespeed shunt, purple/high battery-negative side and grey/low SmartShunt/Lynx system side;
- alternator temperature sensor;
- battery temperature sensor if used by profile/config;
- yellow/green CAN connector protected as unused unless a compatible CAN/BMS integration is later added;
- CAN/app/config access as needed.

Wire-extension guidance from Wakespeed:

- VBat sense: extend with 14 AWG .
- Shunt sense: use twisted pair / instrument cable; shield one end only if shielded.
- Alt+ / Alt-, field/stator: 14 AWG , or 12 AWG if over 20 ft .
- Temperature sensors: keep at least 4 in from noise sources; shielded instrument cable is preferred for extensions.
- Ignition/Feature-In: 14 AWG or 16 AWG .

Build-specific control logic:

- Upfitter #3 OFF : WS500 brown not energized; alternator should not be commanded to charge.
- Upfitter #3 ON : WS500 enabled; charging can occur if engine is running, profile allows charge, and no faults/limits stop it.

- Use Upfitter #3 OFF as the normal emergency/manual charge-disable action.

Do not treat WS500 Feature-In as solved yet:

- Phase 1 leaves white Feature-In reserved.
- Future use may be forced float/standby/charge-disable if a reliable signal is added.
- With current Dumfume internal-BMS/no-CAN batteries, there is no confirmed BMS warning signal to feed Feature-In before disconnect.

WS500 bench-configuration gate

Treat this as a separate de-energized bench session, not the first-engine-run task. Leave Upfitter #3 OFF and the alternator field disabled; remove the WS500 cover only as directed by the current Wakespeed manual, protect the board from static/metal debris, and connect the purchased USB A-to-B/printer-style cable to the documented internal USB port. Establish communications, record firmware/hardware identity, read and save/export the existing configuration before changing anything, then enter and independently verify the build-specific values below. Replace the cover and complete a second read-back before field enable. Do not guess a canned profile or blindly apply the 25% field derate—the Mechman field specification controls.

Phase 4 — WS500 profile/configuration checks

Before first engine run with Upfitter #3 ON :

- Confirm firmware/app access.
- Confirm system voltage detected/configured as 48V / 51.2V nominal class.
- Confirm battery bank capacity reflects 3x 100Ah in parallel (300Ah), not one 100Ah battery.
- Confirm battery charge voltage/current limits are conservative for Dumfume manual values and internal-BMS/no-CAN risk.
- Confirm alternator max-current/current-limit settings are conservative for first run.
- Consider using DIP #8 / small-alternator mode or equivalent conservative field/current limit for first commissioning.
- Confirm field-voltage derate if Mechman alternator uses a 12V field driven from a 48V source. Wakespeed's configuration guidance says many 48V deployments use a 0.25 / 25% normal derate when a 12V field is supplied from 48V ; do not apply or remove this blindly without confirming Mechman field spec.
- Confirm shunt ratio 500A/50mV , Shunt at Battery , common-negative polarity, and positive charging-current sign before trusting readings.
- Confirm alternator temperature sensor reads plausibly.
- Confirm battery temperature sensor reads plausibly or that the profile is explicitly safe without it.
- Confirm Feature-In behavior is known, even if unused.
- Confirm brown enable OFF actually drops/blocks field output.

Suggested conservative first-run intent:

- Do not chase maximum alternator output on day one.
- Prove belt, wiring polarity, sensor readings, regulator control, and shutdown first.
- Increase charge limits only after the basic system behaves predictably.

Phase 5 — first startup / first charge test

Pre-start checklist:

- House bank charged with shore charger and stable.
- Battery temperatures within allowed charging range.
- Main 48V disconnect closed.
- F-04 installed and correct.
- APM-48 installed and green/healthy when live.
- Alternator positive and negative cables landed, torqued, covered, and strain-relieved.
- Negative-shunt bypass proof passed: with the alternator negative return disconnected, Lynx negative showed no stable low-resistance continuity to chassis; reconnecting only through the shunt establishes the intended path.
- WS500 configured and app/monitor available.
- Upfitter #3 OFF before engine start unless deliberately testing enable.
- Multimeter/clamp meter ready.
- Abort plan understood: Upfitter #3 OFF first.

First run sequence:

1. Start truck with Upfitter #3 OFF .
2. Observe belt path, noise, vibration, and any mechanical interference.
3. Verify no unexpected 48V charge voltage/current with WS500 disabled.

4. Turn Upfitter #3 ON to enable WS500 only when ready to observe.
5. Watch WS500 state, battery voltage, alternator current, alternator temperature, battery temperature, and APM LED.
6. Verify charging begins in a controlled way, not an uncontrolled voltage/current jump.
7. Turn Upfitter #3 OFF ; verify alternator current falls near zero.
8. Only after disable behavior is proven, continue with Mechman's loaded test.

Mechman loaded test:

1. Apply a modest electrical load.
2. Raise RPM to 2500 .
3. Measure voltage directly at the 48V battery bank.
4. Verify voltage rises at least 1V over resting voltage.
5. Measure charge-path voltage drop under load: target <0.1V .
6. Measure ground/return-path voltage drop under load: target <0.1V .
7. Stop test if temperature, noise, belt behavior, current sign, voltage, or WS500 faults look wrong.

What to log after first successful run:

- battery resting voltage before start;
- battery voltage during charge;
- alternator current at idle and at test RPM;
- alternator temp trend;
- battery temp;
- charge-path voltage drop;
- return-path voltage drop;
- WS500 profile name/settings summary;
- any WS500 warning/error codes;
- whether Upfitter #3 OFF reliably stopped charge current.

Normal operation

Start/charge:

1. Confirm 48V system is connected and batteries are in a charge-allowed temperature/SOC range.
2. Start engine.
3. Turn Upfitter #3 ON only when alternator charging is desired.
4. Watch early charge behavior after each major change or long idle period.

Stop charging:

1. Turn Upfitter #3 OFF .
2. Verify alternator charge current falls near zero if monitoring is available.
3. Leave main 48V disconnect closed while engine is running unless the alternator is confirmed inactive.

Full shutdown/service:

1. Upfitter #3 OFF .
2. Wait for charge current to collapse.
3. Stop engine if practical.
4. Open main 48V disconnect / service disconnects.
5. Then work on cables/fuses/components using insulated tools and PPE.

Fault response

If any battery BMS trip, over-voltage, unexpected disconnect, APM alarm, belt issue, high alternator temp, smoke/smell, or weird noise occurs:

1. Upfitter #3 OFF immediately.
2. Do not open the main 48V disconnect while charge current is active unless there is a more immediate hazard.
3. Let alternator current fall.
4. Stop engine if needed.
5. Inspect APM LED/chirp status.
6. Inspect fuses, lugs, belt, and cable strain relief.
7. Do not re-enable until cause is understood.

Specific symptom mapping:

- No charge: confirm Upfitter #3 , F-15 , brown wire voltage, WS500 state, field harness/polarity, sense wiring, shunt sign, and profile.
- Low charge: check belt slip, current limit/profile, alternator temp limit, voltage sense location, battery SOC, and high-resistance connections.
- Squeal: belt path/tension/alignment issue.
- Howl: Mechman flags grounding or battery problem as possible cause.
- Growl: possible bearing/mechanical issue.
- APM red LED or chirp: replace APM; do not treat it as healthy.
- Unexpected battery disconnect: treat as serious; reduce charge voltage/current and investigate BMS state before any further alternator charging.

Internal-BMS/no-CAN operating policy

This build does not currently have a confirmed Wakespeed-compatible BMS communication path from the Dumfume batteries.

Therefore:

- WS500 profile should be conservative.
- Alternator current limit should not assume perfect current sharing or BMS communication.
- Battery BMS opening under alternator charge is not a normal control strategy.
- APM-48 is backup protection, not permission to ignore disconnect risk.
- Avoid alternator charging when batteries are near full, below low-temp charge threshold, overheated, faulted, or behaving abnormally.
- If a reliable future BMS/relay/charge-allowed signal is added, revisit Feature-In or other WS500 shutdown integration.

Open items before final commissioning

- Confirm exact Mechman alternator field polarity and installed WS500 harness type (PH vs NH).
- Confirm whether the Mechman 48V alternator field is 12V or true 48V for derate/profile setup.
- Confirm alternator negative/case behavior for APM black-lead landing and prove there is no second negative/chassis path around the dedicated negative-return shunt.
- Confirm final WS500 profile values against Dumfume manual and Mechman/Wakespeed support guidance.
- Record final measured alternator-to-bed route lengths for the high-current pair and WS500 alternator-leg extensions.
- Record first-run measurements in logs/LOG.md after commissioning.

One-page shop checklist

Before mechanical install:

- ☐ Official Mechman bracket PDF open for belt routing.
- ☐ Owner-confirmed 7.3L single-generator routing image available for orientation: A closest to engine, B furthest from engine.
- ☐ Batteries disconnected as required for work.
- ☐ Eye protection / insulated tools / no jewelry.
- ☐ Alternator terminals identified.
- ☐ WS500 harness type known.

Mechanical install:

- ☐ Bracket installed.
- ☐ Idler installed.
- ☐ Saddle alternator installed.
- ☐ T-mount alternator installed.
- ☐ 88.06 in belt installed per diagram.
- ☐ Belt alignment/tension/clearance checked.
- ☐ Hardware torqued to OEM/Mechman/Ford spec.

Electrical install:

- ☐ APM red to B+ .
- ☐ APM black to B- or case bolt per alternator ground type.
- ☐ APM terminals not under battery cable lugs.
- ☐ Alternator positive 2/0 routed/protected to F-04 / Lynx.

- ☐ Alternator dedicated negative 2/0 routed directly to Lynx/system negative.
- ☐ Wakespeed shunt installed on the battery side of the hard-mounted SmartShunt in the common battery-negative path; purple on battery-negative side, grey on SmartShunt/Lynx system side; no return bypasses either shunt.
- ☐ F-04 200A/80V installed at house end.
- ☐ Cable covers/boots/strain relief installed.

WS500:

- ☐ Brown enable from Upfitter #3 through F-15 3A .
- ☐ PH-VAN short red combined power/positive-sense lead through one F-12/F-13-PHVAN 15A/80VDC fuse in a compatible holder at the house/main positive bus; do not extend.
- ☐ Negative sense correct.
- ☐ Shunt current sense correct and sign checked.
- ☐ Alternator temp sensor installed.
- ☐ Battery temp handling confirmed.
- ☐ Profile configured for 48V / 300Ah bank.
- ☐ Conservative current/field limit for first run.

First run:

- ☐ Upfitter #3 OFF at engine start.
- ☐ Belt/noise check passes.
- ☐ No unexpected charge with WS500 disabled.
- ☐ Upfitter #3 ON only under observation.
- ☐ Upfitter #3 OFF verified to stop charge current.
- ☐ 2500 RPM loaded test completed only after basic control works.
- ☐ Charge and ground path drops each $<0.1V$ under load.
- ☐ APM green/healthy after test.
- ☐ Measurements logged.